

Thermo Scientific Multidrop Combi Remote Control Command Sets

Rev. 1.4



Thermo Scientific Multidrop Combi

Remote Control Command Sets

Rev. 1.4

Copyright © 2009 Thermo Fisher Scientific. All rights reserved. Printed in Finland. First edition published in 2005. Reproduction of the accompanying user documentation in whole or in part is prohibited.

The Multidrop Combi is protected by the following patent: US 4995432.

“Microtiter” and “Multidrop” are registered trademarks of Thermo Fisher Scientific.

All other trademarks and registered trademarks are the property of their respective holders.

Thermo Fisher Scientific reserves the right to change its products and services at any time to incorporate technological developments. This manual is subject to change without prior notice as part of a continuous product development. Although this manual has been prepared with every precaution to ensure accuracy, Thermo Fisher Scientific assumes no liability for any errors or omissions, nor for any damages resulting from the application or use of this information. This manual supersedes all previous editions.

Thermo Fisher Scientific shall not be liable for any damages whatsoever arising out of the use or inability to use this product.

About This Document

Intended users

This document is written for the integrator and provides information on the Thermo Scientific Multidrop Combi remote control.

Read the document in its entirety before operating the instrument.

How to use this document

This document is designed to give you the information to:

- Use the Multidrop Combi basic commands
- Use the Multidrop Combi advanced commands
- Use the Multidrop Combi emulated commands

For more information

For the latest information on products and services, visit our worldwide websites on the Internet at:

<http://www.thermo.com>

<http://www.thermo.com/multidrop>

In an effort to produce useful and appropriate documentation, we appreciate your comments on this document to your local Thermo Fisher Scientific representative.

About This Document

For more information

Contents

	Intended users	3
	How to use this document	3
	For more information	3
Chapter 1	Introduction	9
	Purpose of the document	9
	Overview	9
Chapter 2	Communication Interfaces	11
	RS232	11
	USB	11
	Sending commands to the instrument	12
	Receiving responses from the instrument	12
	Abort	13
	Communication timeouts	13
Chapter 3	Command Set	15
	Command format	15
	Response format	15
	Buffering	16
	User commands	16
	Abort	17
	BGN – start a protocol from memory	17
	DIS – dispense plate	18
	EAK – error acknowledge	19
	EMP – empty dispenser hoses	22
	PLA – define remote plate	22
	POU – move the plate to the out position	24
	PRI – prime dispenser hoses	24
	QIT – quit the computer control	25
	RIR – read RFID information from cassette	25
	RIW – write RFID information to cassette	25
	RST – restart instrument	26
	SCT – set remote cassette type	26
	SCV – set remote column volume	26
	SDH – set remote dispensing height	27
	SDO – set remote dispensing order	27
	SHA – shake	28
	SOF – set remote dispense offset	29
	SPL – set remote plate type	29
	SPS – set pump speed	30
	SPV – set remote dispensing predispense volume	30
	TIM – set date and time	30

	VER – version request	31
	Service commands	31
	BOOT – enter bootstrap load mode	32
	GET – report user defined parameters	32
	HEX – send a hex file record	33
	LOG – send error log	33
	MOP – move pump	36
	MOX – move plate carrier	36
	MOY – move Y	36
	MOZ – move Z	37
	PAR – set nonvolatile parameters	37
	PPA – set remote pump move parameters	38
	PUT – save user-defined parameter to nonvolatile memory	38
	REA – read a byte or word value	39
	REP – report	40
	RFS – restore factory settings	44
	SRE – status report	44
	WCA – write cassette	46
	WRI – write a byte or word value	47
	WPL – write plate	47
	WPR – write protocol	49
Chapter 4	Emulated Command Set	53
	Commands	53
	Responses and error codes	55
	Default protocols	56
Chapter 5	Using the Commands	59
Appendix A	Plate Definitions	61

Figures

Figure 3–1. Four arguments of the command 23

Figure 3–2. Distance arguments of the command 49

Figure A–3. Plate distance definitions 61

Tables

Table 3–1. Error codes..... 20

Table 3–2. Remote plate definitions..... 23

Table 3–3. Error codes..... 35

Table 3–4. Truth table for changing the attributes of an existing protocol.. 52

Table 4–5. Default values and limits in the two emulation modes 55

Table A–6. List of predefined plates 61

Chapter 1

Introduction

Purpose of the document

This document is the serial interface specification of the Multidrop Combi device.

Overview

The Multidrop Combi microplate dispenser is intended for professional research use by trained personnel. The instrument is intended for automated dispensing of up to eight different reagents simultaneously into a variety of microplates and strips in 96-, 384- or 1536-well plate format. Use for self-testing is excluded.

It will be an additional or substitution unit to the Thermo Scientific Multidrop 384, Thermo Scientific Multidrop Micro and Thermo Scientific Multidrop DW line and will also provide new features by dispensing smaller volumes, accepting a variety of Thermo Scientific Microtiter plates and strips in 96-, 384- and 1536-well plate format.

Introduction

Overview

Chapter 2

Communication Interfaces

The instrument has two alternative computer interfaces: RS232 and USB.



Note It is recommended to use USB to improve the reliability of traffic in robot applications. ▲

The interface to use is selected from the Multidrop Combi user interface. It is only possible to connect to the instrument when the instrument is idle (no work is done through the user interface). To connect to the instrument the computer must send a version query (command VER) or, if an emulation mode is selected, any command.

To disconnect from the remote mode the computer must send the QIT command to the instrument or, in emulation mode, the Q command. In either case the remote control can also be terminated by pressing the **STOP** button on the keyboard.

RS232

The serial connector on the instrument is a 25-pin male D connector. The reception is through pin 2, transmission through pin 3 and the signal ground at pin 7.

The port setup for the RS232 interface is: baud rate 9600, 1 start bit, 8 data bits, 1 stop bit and no parity.

XON/XOFF software handshaking is used. The XON character is 0x11 and the XOFF character 0x13. The maximum length of a command line is limited by the receive buffer size, which is 512 bytes. Note, however, that XOFF will be sent when the buffer is $\frac{3}{4}$ full (384 characters).

No frames or checksums are used in the RS232 protocol. The instrument ignores any NUL bytes (0x00) it receives.

USB

For connecting using the USB port of the PC, the operating system of the PC must have USB support. The physical connection on the instrument is a standard USB series “B” receptacle. The instrument complies with the USB specification 2.0 full-speed device.

The following identification information is returned during enumeration:

Vendor id:	0x0AB6
Product id:	0x0344
Manufacturer:	Thermo Fisher Scientific Oy
Product descriptor:	The name stored in nonvolatile parameters, default is Multidrop Combi.
Serial number:	The serial number string of the instrument.

The instrument has one configuration, which is numbered 1, and in this configuration one interface. This interface contains a bulk IN endpoint (0x81) and a bulk OUT endpoint (0x01). The instrument receives commands through the OUT endpoint and sends responses through the IN endpoint.

Sending commands to the instrument

In order not to consume USB bandwidth when the instrument is unable to receive data, the PC application should first ask the instrument how much data it can receive. This is done by a vendor request to the control endpoint. The Setup packet fields of this request must be set as follows:

bmRequestType	0xC0
bRequest	0x02
wValue	0x00
wIndex	0x00
wLength	0x02

The instrument returns two bytes of data to this request. The first byte is the LSB and the second the MSB of the number of characters the instrument can receive through the bulk OUT endpoint. If the PC application has more data to send than the instrument can receive, the application should send as much data as the instrument can receive and then poll the instrument at regular intervals to find out when the instrument can receive more data.

Receiving responses from the instrument

To receive data from the instrument the PC application must first ask how much data the instrument has to send and then read exactly that much data from the bulk IN endpoint. The query is done by a vendor request to the control endpoint. The Setup packet fields of this request must be set as follows:

bmRequestType	0xC0
bRequest	0x01
wValue	0x00
wIndex	0x00
wLength	0x02

The instrument returns two bytes of data to this request. The first byte is the LSB and the second the MSB of the number of characters the instrument has waiting to be read by the PC application. If the returned value is nonzero, the PC application must read exactly that much data from the bulk IN endpoint before sending another query to the control endpoint. The instrument only sends data after it has received the query and has responded with a nonzero value. It then sends exactly the amount of data it reported. If the instrument reports no data, the PC application must keep sending the query at regular intervals to become aware of any new data available.

Abort

Because there is no way the PC application can force an Abort to the bulk OUT endpoint past previous commands, aborting must be performed in two phases. First the PC must send the following vendor request to the control endpoint:

bmRequestType	0x40
bRequest	0x00
wValue	0x00
wIndex	0x00
wLength	0x00

Then the PC must send the Abort command in a normal manner through the bulk OUT endpoint.

The vendor request causes the instrument to discard all commands it may already have received. The instrument then reads the bulk OUT endpoint, discarding all received commands until it receives the Abort command.

Communication timeouts

Whenever the instrument has something to send, but fails to send anything for 30 seconds, it is considered a fatal communication error and the instrument is reset.

Communication Interfaces

Communication timeouts

Chapter 3

Command Set

This chapter describes the commands and the responses as the controlling software and the instrument see them without the framing added by the data transport layer.

All commands are ASCII lines terminated with either a <CR> or a <LF> character.

The length of a single command should be kept at 350 characters or less. Longer commands may cause a deadlock on the RS232 port due to flow control. The same limit should be honored even when the USB port is used.

A CR or LF preceded by a CR or a LF is an empty command and is ignored by the instrument.

Responses from the instrument are ASCII lines terminated with a CR followed by a LF.

Command format

With a few exceptions, the commands comprise a 3-character command identifier followed by command parameters applicable to the command. The identifier itself and any ASCII hex command parameters are not case sensitive, but the rest of the parameters are. One or more <SPACE> characters separate the fields of the command. Trailing spaces are ignored.

Example: VER<CR><LF>
or ver<CR><LF>

In some cases it is not necessary to supply all command parameters.

Response format

Response lines start with the same 3-character identifier as the command being executed. The identifier in the responses is always in uppercase.

The first response line only contains the command identifier and it is sent when the execution of the command starts.

The last response line contains the command identifier, end field characters 'END' and the status field, ui_status, which is '0' for a successful completion, otherwise an error or a warning code. See the EAK command for possible values that the status field may have.

Between the first and the last response lines there are command-specific data lines if data is returned. Note that these lines may not be there if an error is reported.

Buffering

The instrument is able to receive more than one command line at a time.

The flow control of the communication protocol ensures that the receive buffer never overflows. The instrument processes the commands in the order they were received with the exception of the Abort command, which is processed immediately after it has been received.

The instrument also buffers the response lines. If the PC is slow in processing the responses, the flow control of the communication protocol prevents the receive buffer of the PC from overflowing. Due to response line buffering the instrument continues processing the commands until its transmit buffer becomes full.

User commands

The command arguments and response data in the command descriptions have a prefix describing the type of argument/data:

- a_ ASCII string (non-numeric).
- sc_ Signed character (-128 to 127).
- uc_ Unsigned character (0 to 255).
- si_ Signed 16-bit integer (-32768 to 32767).
- ui_ Unsigned 16-bit integer (0 to 65535).
- sl_ Signed long (-2147483648 to 2147483647).
- ul_ Unsigned long (0 to 4294967295).

Since the white space character is used as a field separator in the commands and responses, an ASCII string argument or response data may not contain white space. Therefore, the ampersand character '&' is used in the ASCII string arguments and response data to represent the white space character. The instrument converts the & character in the commands to white space, and a white space in the responses is represented with the & character.

If no other limits to a command argument or response data are given, it may have any value that the data type permits.

Arguments and data in square brackets [] are optional.

The timing resolution of the instrument is 10 ms. All timing arguments for the commands are given in 1/100 seconds, which is in accordance with the timing resolution.

For clarity, the <CR><LF> characters at the end of a command or response line are not shown in the following descriptions.

Abort

This command is an exception to the rule of a 3-character command length. Abort is the single ASCII control character ESC. This command aborts execution of the current command in the instrument. The Abort command is always processed before any other commands in the instrument's receive buffer and it flushes the receive buffer. Abort stops any dispensing, the plate is driven out and the pump up. Abort clears any pending error, but an error during abort is reported. Further Abort commands during the execution of abort are ignored.

Command: <ESC> (Esc character: decimal 27, hex 1B)

Response: ABO
ABO END ui_status

Example: <ESC>
:
ABO
ABO END 0

Note that the responses to the Abort command use the three-letter mnemonic ABO.

The remote operation can also be aborted from the keyboard. The computer detects this by receiving an unexpected ABO response. However, in emulation mode the instrument just stops responding.

BGN – start a protocol from memory

This command starts the protocol currently in the user interface or a protocol stored in the nonvolatile memory.

Command: BGN [ui_plate_type a_protocol_name]

ui_plate_type 0..29.

The following plate types are factory defined (see Appendix A: "Plate Definitions"):

96 standard (15mm)	0
96 low DW (22mm)	1
96 DW (44mm)	2
384 low vol. (7.5mm)	3
384 low prof. (10mm)	4
384 standard (15mm)	5
384 DW (22mm)	6
384 DW (44mm)	7
1536 low vol. (5mm)	8
1536 std (10.5mm)	9

More plate types may be defined with the WPL command. This argument is optional. If the plate type is given, the protocol name must also be given.

a_protocol_name Name of a protocol stored in the nonvolatile memory of the instrument. This argument is optional and if given, the plate type must also be given.



Note The maximum length of the name is 20 characters. The allowed characters are as follows: 'A'...'Z', 'a'...'z', '0'...'9', '-', '_', and '&' which translates into white space. ▲

If the plate type and protocol name are not given, the protocol currently in the user interface is run.

The protocol started with the BGN command overwrites the current remote control protocol.

Response: BGN
BGN END ui_status



Note The protocol overwrites all settings done for the remote control protocol, i.e., if a DIS command is executed after a BGN command, exactly the same protocol is run as with the BGN command. ▲

DIS – dispense plate

This command dispenses the plate. Parameters for the dispensing should have been set up with the commands PLA, SDO, SCT, SCV, SDH, SOF, SPL, SPP, SPV and SSC prior to dispensing. The pump must have been primed either manually with the **PRIME** button or with the PRI command. The priming vessel and the rotor cover must be in place when dispensing.

Command: DIS
Response: DIS
DIS ui_col ui_row
:
DIS END ui_status
ui_col Dispensed wells column address, 1...
ui_row Lowest number row dispensed, 1..4. Since the instrument is an eight-channel dispenser, seven other rows were also dispensed. For a 96-well plate, ui_row is always 1 and all rows 1..8 are dispensed simultaneously. For a 384-well plate, ui_row is 1 or 2 and all even-numbered rows or all odd-numbered rows are dispensed. For a 1536-well plate, ui_row is 1..4 and every fourth row is dispensed, starting from the row in the response.

Example: DIS
:
DIS
DIS 1 1
:
DIS END 0

In this example the first dispensed column is column 1 and wells in the rows A..H were dispensed.

Example: *DIS*
:
DIS
:
DIS 3 4
:
DIS END 0

In this example the dispensed column is column 3 and the lowest number row is row D, so the plate must be a 1536-well plate.

EAK – error acknowledge

This command must be used to clear an error status from the instrument. When the instrument detects an error, it reports it in the END response of the current command or the first command after startup. When an error is detected, the current operation is terminated. The instrument may not be in a coherent state after an error, so it is best to abort the current run and restart from the beginning after the problem is fixed. In error state the instrument executes only the commands EAK, VER, BOOT and Abort. Other commands are acknowledged, but they immediately return with the current error code.

Command: *EAK*
Response: *EAK*
EAK END 0

Error codes above 1000 in the error log:

A number of error codes above 1000 may appear in the error log. These are fatal errors and cause the firmware to halt. See the LOG command description for more details.

Error codes reported in response status are presented in Table 3–1.

Table 3–1. Error codes

Code	Explanation	Suggested action
0	The command was executed successfully.	
1	Internal firmware error. When this error occurs, the internal software halts. This error can only be seen in the error log returned for the LOG command.	Contact service.
2	The instrument did not recognize the command it received.	Contact the PC software vendor.
3	The arguments of the received command are not valid.	Contact the PC software vendor.
4	Pump position error.	Reduce dispensing speed or, if using the small tube cassette, use the standard tube cassette instead. If the problem persists even with normal viscosity fluids, contact service.
5	Plate X position error.	If this error is reported repeatedly, contact service.
6	Plate Y position error.	If this error is reported repeatedly, contact service.
7	Z position error.	If this error is reported repeatedly, contact service.
9	Attempt to reset the serial number.	Do not try to reprogram the serial number.
10	Nonvolatile parameters lost.	Contact service.
11	No more memory for storing user data.	Release some user data no longer needed.
12	Pump or X motor was running.	Do not attempt to perform any other operation than stopping a continuous prime/empty/shake when such a continuous run is ongoing.
13	X and Z positions conflict.	Contact service.
14	Cannot dispense when pump not primed.	Prime the pump before dispensing.
15	Missing prime vessel.	Insert the priming vessel.
16	Rotor shield not in place.	Slide the rotor cover in place.
17	Dispense volume for all columns is 0.	Select a nonzero volume for at least one plate column.
18	Cannot dispense because the plate type is invalid (bad plate index).	Use a valid plate index (SPL command).
19	Cannot dispense because the plate has not been defined.	Define the plate (WPL command).
20	Invalid rows in plate definition.	Check the WPL or PLA command.

Continued

Cont.

Code	Explanation	Suggested action
21	Invalid columns in plate definition.	Check the WPL or PLA command.
22	Plate height is invalid.	Check the WPL or PLA command.
23	Plate well volume is invalid (too small or too big).	Check the WPL or PLA command.
24	The cassette type is invalid (bad cassette index).	Use a valid cassette index (SCT command).
25	Cassette not defined.	Define the cassette (WCA command).
26	Volume increment defined for the cassette is invalid.	Check the WCA command.
27	Maximum volume defined for the cassette is invalid.	Check the WCA command.
28	Minimum volume defined for the cassette is invalid.	Check the WCA command.
29	Minimum or maximum pump speed defined for the cassette is invalid.	Check the WCA command.
30	Pump rotor offset in the cassette definition is invalid.	Check the WCA command.
31	Not used.	Not used.
32	Dispensing volume is not within limits allowed for the cassette.	Correct the volume or use a different cassette.
33	Invalid selector channel.	Correct the selector channel number.
34	Invalid dispensing speed.	Use a valid value.
35	Dispensing height is too low for the plate.	Increase dispensing height of the protocol.
36	Tips' predispense volume is not within limits for the cassette.	Change the predispense volume of the protocol.
37	Invalid dispensing order of the plate.	Use dispensing order 0 or 1.
38	Invalid X or Y dispensing offset.	Fix the offsets. Note that the limits depend on the used plate type.
39	RFID option not present.	Purchase the RFID option for the instrument.
40	RFID tag not present.	Insert a cassette type containing a RFID tag.
41	RFID tag data checksum is incorrect.	If the cassette has never been calibrated, calibrate it. Else discard the cassette.
43	Wrong cassette type.	Use a cassette type matching the cassette type of the current protocol.
44	Protocol or plate may not be modified or deleted because it is currently used.	This error may be returned in response to the WPL and WPR commands. See the command descriptions for the limitations causing this error.
45	Cannot modify or delete protocol,	This error may be returned in response to

Continued

Cont.

Code	Explanation	Suggested action
	plate or cassette because it is read only.	the WCA, WPL and WPR commands. See the command descriptions for the limitations causing this error.

EMP – empty dispenser hoses

This command is used for emptying the dispenser hoses remotely from the computer.

Command: EMP ul_volume [ui_mode]

ul_volume This is the emptying volume in 1/10 µl. The upper limit is 100000 (10 ml). The lower limit depends on the current cassette and is 100 (10 µl) for the standard tube cassette and 10 (1 µl) for the small tube cassette. If ui_mode is used, this is ignored, but must still be within the valid range.

ui_mode 0 = continuous emptying OFF
1 = continuous emptying ON.
If this optional parameter is used, the volume is ignored. If continuous emptying is switched on, the next command should switch continuous emptying off. Also other commands are possible as long as they do not cause any of the motors to run. If a position error is detected during continuous emptying, the emptying stops. The error code is returned to the next received command.

Response: EMP
EMP END ui_status

PLA – define remote plate

This command defines the physical dimensions of a plate for the remote control. After startup the default is a 15 mm high standard 96-well plate. Note that command SPL could also be used to select the remote control plate from one of the plates types stored in nonvolatile memory.

Command: PLA ui_first_column_x ui_last_column_x ui_first_row_y ui_last_row_y ui_columns ui_rows ui_plate_height ui_max_well_volume

ui_first_column_x Distance from the left edge of the plate to the center point of the wells in the first column.

ui_last_column_x Distance from the left edge of the plate to the center point of the wells in the last column.

ui_first_row_y Distance from the front edge of the plate to the center point of the wells in the first row.

si_last_row_y Distance from the front edge of the plate to the center point of the wells in the last row.

ui_columns Number of well columns on the plate, 1...48.

ui_rows Number of well rows on the plate, 8, 16 or 32.

ui_plate_height Height of the plate 0...5000.

ui_max_well_volume Maximum volume to dispense into a well in

1/10 µl, 1...25000. Note that even though this range is accepted by the PLA command, an error is reported during dispensing if a column volume is outside the limits of the used cassette.

Response: PLA
PLA END ui_status



- Note**
- 1) All distances in the command are in 1/100 mm (Table 3–2).
 - 2) The distance between the first row and the last row must be 63 mm ± 1 mm for 8 rows, 67.5 mm ± 0.5 mm for 16 rows and 69.75 mm ± 0.1 mm for 32 rows.
 - 3) The origin of the coordinates is at the front left corner of a plate inserted into the plate carrier of the instrument.
 - 4) The limits depend on the number of rows on the plate. There must always be at least 1/3 of the row-to-row distance to a user position limit from the center point of any well (see commands MOX and MOY), except that the X position lower limit is 0. ▲

Table 3–2. Remote plate definitions

Row-to-row	ui_first_column_x		ui_last_column_x		ui_first_row_y		ui_last_row_y	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
9mm	300	11550	300	11550	(7300)	(7675)	1000	1375
4.5mm	150	11700	150	11700	(7600)	(8275)	850	1525
2.25mm	75	11775	75	11775	(7750)	(8575)	775	1600

The values in parentheses assume the distance between the first and the last row to be the nominal one given in Note 2) above.

Figure 3–1 shows how the first 4 arguments of the command are defined.

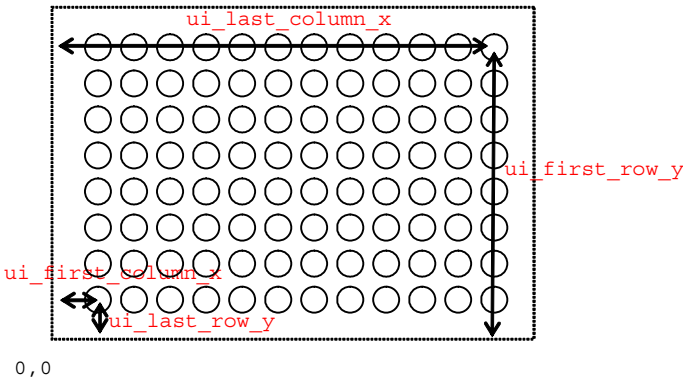


Figure 3–1. Four arguments of the command

Appendix A: "Plate Definitions" contains a list of predefined plates.

POU – move the plate to the out position

This command moves the plate to the out position where the plate clamp is released.

Command: POU
Response: POU
POU END ui_status

PRI – prime dispenser hoses

This command is used for filling the dispenser hoses with dispensing solution remotely from the computer.

Command: PRI ul_volume [ui_mode]

ul_volume This is the priming volume in 1/10 µl. The upper limit is 100000 (10 ml). The lower limit depends on the current cassette and is 100 (10 µl) for the standard tube cassette and 10 (1 µl) for the small tube cassette. If ui_mode is used and is not 2, this is ignored, but must still be within the range. Note that priming forces the rotor to stop at the offset angle defined for the current cassette, so the actual priming volume may be higher than given in the command. Also, if the pump is not already primed, the counting of the volume only starts after the pump home position is established.

ui_mode 0 = continuous priming OFF.
1 = continuous priming ON.
2 = volumetric calibration volume priming.
3 = volumetric calibration continuous priming ON.
If this optional parameter is 0, 1 or 3, the volume is ignored. If continuous priming is switched on, the next command should switch continuous priming off. Also other commands are possible as long as they do not cause any of the steppers to run. If a position error is detected during continuous priming, the priming stops. The error code is returned to the next received command. If volumetric calibration mode 2 is selected, the pump must already be primed and the plate must be out. No other stepper than the pump is run in volumetric calibration modes 2 and 3.

Response: PRI
PRI END ui_status

The priming vessel and the rotor cover must be in place. The plate carrier will automatically be driven out of the way if necessary when priming. Before dispensing is possible, priming must be done either manually from the user interface or with this command in the following cases:

- After instrument startup.

- After the pump has been emptied either manually from the user interface or with the EMP command.
- After the rotor cover has been out of place.
- After changing the selector channel.

QIT – quit the computer control

This command ends the computer control.

Command: QIT
Response: QIT
QIT END ui_status

Do not send a VER command until after the END response for this command is received. The QIT command flushes the receive buffer and a VER sent immediately after QIT may not be recognized.

RIR – read RFID information from cassette

This command can be used to read identification of the current cassette.

Command: RIR

Response: RIR
RIR a_calibration a_serial ui_used_cal ui_used_total
RIS END ui_status

a_calibration Date and time of cassette calibration, same format as in the TIM command.

a_serial Cassette serial number.

ui_used_cal Cassette usage indication at time of last calibration. This is the ui_used_total value at the time of calibration.

ui_used_total Total cassette usage indication. 0 for an unused cassette, increasing to 100 when the specified cassette lifetime has been reached. If the cassette is still used, the value keeps increasing.

RIW – write RFID information to cassette

This command can be used to write calibration information to the current cassette.

Command: RIW a_calibration [a_serial]

a_calibration Date and time of cassette calibration, same format as in the TIM command. If a_serial is not given, the ui_used_cal counter is set to the current value of the ui_used_total counter (see RIR command).

a_serial Cassette serial number. This is optional and should only be used when the cassette is calibrated for the first time. When a_serial is given, both ui_used_cal and ui_used_total counters are reset (see RIR command). The maximum length of the serial number string is 20 characters.

Response: RIW
RIW END ui_status

RST – restart instrument

This command restarts the instrument. The effect is the same as switching the instrument power off and then back on.

Command: RST
Response: RST
RST END 0

The instrument waits until all data to the computer is sent before executing a software reset (restart).

SCT – set remote cassette type

This command sets the cassette type for the DIS command. After startup the default is type 0 (Standard tube cassette).

Command: SCT ui_type

ui_type 0...3
Types 0 and 1 are factory defined:
0 = Standard tube cassette
1 = Small tube cassette
Cassettes 2 and 3 may be defined with the WCA command.

Response: SCT
SCT END ui_status

SCV – set remote column volume

This command selects columns and sets column dispensing volumes for the DIS command. The volumes are used in the subsequent dispense commands.

Command: SCV ui_column ui_volume

ui_column 0 = set to all columns.
1...48 = set to given column.
You can set the volume for 48 columns even if the current plate type does not have that many columns.
ui_volume 0 = do not dispense to this column.
1...25000 dispensing volume in 1/10 µl.
Even though this range is accepted by the SCV command, an error is reported to the DIS command if the cassette type does not allow the selected volume(s).

Response: SCV
SCV END ui_status

Example: SCV 0 500

This command selects all columns and sets the column dispensing volume to 50 µl.

Example: *SCV 0 0*
SCV 1 100
SCV 3 200
SCV 5 300
SCV 7 400

This command unselects all columns and then selects columns 1, 3, 5 and 7 to be dispensed using 10, 20, 30 and 40 µl.

SDH – set remote dispensing height

This command sets the dispensing height for the DIS command.

Command: *SDH ui_dispensing_height*

ui_dispensing_height Distance from the plate bottom to the lowest part of the dispensing cassette when dispensing. This is in 1/100 mm and may range from 500 to 5500. Note that the dispensing tips may be slightly higher depending on the cassette type. When dispensing is started, an error is reported if the dispensing height is less than the current plate height + 100.

Response: *SDH 1700*
SDH END ui_status

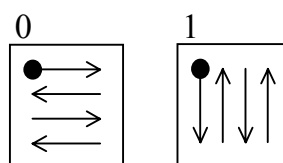
For example, if the plate height is 15 mm, setting a dispensing height of 1700 leaves a 2 mm gap between the cassette and the top of the plate.

SDO – set remote dispensing order

This command selects the dispensing order for the DIS command: whole row first or whole column first. After startup the default is row wise.

Command: *SDO ui_order*

ui_order Value 0 dispenses row wise and 1 column wise. The figures below illustrate the corresponding filling sequences.



For an 8-row plate the selection of the dispensing order makes no difference.

Response: *SDO*
SDO END ui_status

SHA – shake

This command is used to shake the plate the given time using the given shake cycle.

Command: SHA ul_time ui_distance ui_speed [ui_mode]

ul_time	Total shaking time in 1/100 seconds.
ui_distance	Distance of the shaking movement 1...5 mm.
ui_speed	Frequency 1...20 Hz.
ui_mode	0 = continuous shaking OFF 1 = continuous shaking ON.

If this optional parameter is used, the time is ignored.
If continuous shaking is switched on, the next command should switch continuous shaking off. Also other commands are possible as long as they do not cause any of the steppers to run. If a position error is detected during continuous shaking, the shaking stops. The error code is returned to the next received command.

Response: SHA
SHA END ui_status

Example: *SHA 500 1 20*
SHA
:
SHA END 0

This example does a 5-second shake at 20 Hz and the distance of 1 mm.

All shake distance/speed combinations are not possible. If the acceleration or speed is too high, an error is reported. The maximum distances for different speeds are listed below:

Speed [Hz]	Distance [mm]
1	5
2	5
3	5
4	5
5	5
6	5
7	5
8	5
9	4
10	3
11	3
12	2
13	2
14	2
15	1
16	1
17	1
18	1

Speed [Hz]	Distance [mm]
19	1
20	1

If the plate carrier is in the out position when the shaking is started, it is first moved in enough to clamp the plate to the plate carrier.

SOF – set remote dispense offset

This command sets the amount by which the dispensing position is moved in the X and Y directions from the nominal center point of a well for dispensing in remote operations. It is used for adjusting the dispensing position to the center of a well for plates with slightly different dimensions, or for deliberately offsetting the dispense position from the center of the well.

Command: SOF si_dispense_x_offset si_dispense_y_offset

Response: si_dispense_x_offset The range is -300..300.
si_dispense_y_offset The range is -300..300.
SOF
SOF END ui_status

The offsets are in 1/100 mm. The offset limits of the command are typical maximum offsets for a 96-well plate. When dispensing is started, the offsets are checked to be no more than $\pm 1/3$ of the row-to-row distance of the plate. If they are more, an error is reported. The limits for a typical 384-well plate are ± 150 and for a typical 1536-well plate ± 75 .

SPL – set remote plate type

This command selects the plate type for the remote operations. Note that command PLA could also be used to define the plate for remote control.

Command: SPL ui_plate_type

Response: ui_plate_type Plate type 0...29.
The following types are factory defined (see Appendix A: "Plate Definitions"):

96 standard (15mm)	0
96 low DW (22mm)	1
96 DW (44mm)	2
384 low vol. (7.5mm)	3
384 low prof. (10mm)	4
384 standard (15mm)	5
384 DW (22mm)	6
384 DW (44mm)	7
1536 low vol. (5mm)	8
1536 std (10.5mm)	9

Plate types 10...29 may be defined with the WPL command.

SPL 6
SPL END ui_status

SPS – set pump speed

This command sets the relative pump speed for dispensing.

Command: SPS ui_speed

ui_speed The top speed used for the pump motor during dispensing, 1...100. This is in % of the speed range defined for the current cassette type (see the WCA command).

Response: SPS 100
SPS END ui_status

The pump has certain absolute minimum and maximum speeds. Each cassette type has limits for the pump speeds allowed for that cassette. For example, if the minimum pump speed allowed for a cassette is 40% and the maximum 80%, command SPS 30 would result in a pump speed of 52%. This result comes from the following calculation:

The requested speed 30% of the cassette speed range corresponds to 12% of the total pump speed range ($0.3 * (80\% - 40\%) = 12\%$). This is added to the minimum cassette speed of 40%, resulting in a pump speed of 52%.

SPV – set remote dispensing predispense volume

This command sets the volume predispensed to the priming vessel before dispensing the plate.

Command: SPV ul_prime_volume

ul_prime_volume The volume to predispense into the priming vessel before dispensing in 1/10 µl. The range is 10...100000. However, note that when the dispensing is started, the volume is rounded up to the nearest number of 1/4 pump revolutions. Also, the different cassettes have different limits for the predispense volume, so the DIS command may return an error even if the SPV command accepts the volume.

Response: SPV 100
SPV END ui_status

TIM – set date and time

This command sets the current date and time in the instrument. The current date and time is reset to 01.01.2000 00:00:00 at every power up and the clock calendar will not start until this command is received. It is not necessary to set the date and time, but it is strongly recommended because many reports as well as the error log include a date and time.

Command: TIM dd.mm.yyyy HH:MM:SS

dd	Day of month, 1...31, but not more than there are days in the month.
mm	Month, 1...12.
yyyy	Year, 2000...9999.
HH	Hour, 0...23.
MM	Minutes, 0...59.
SS	Seconds, 0...59.

Response: TIM
TIM END ui_status

One to four digits are accepted in each of the date and time positions, so, for example, days 1 and 0001 would be valid, but 00001 invalid.

The clock calendar will roll over to 01.01.2000 00:00:00 after 31.12.9999 23:59:59.

Example: TIM 28.08.2001 12:21:35
TIM
TIM END 0

VER – version request

This command must be sent to the instrument to enter the computer control mode. The instrument does not respond to other commands when not in computer control mode. The instrument responds to this command even when waiting for error acknowledgement. See the EAK command.

Command: VER

Response: VER
VER a_instrument ui_major.ui_minor.ui_build a_serial
VER END ui_status

a_instrument	Instrument name stored in the nonvolatile memory. The default is: 'MultidropCombi'.
ui_major	Major version number.
ui_minor	Minor version number.
ui_build	Build number.
a_serial	The serial number string as set by the PAR 0 command. If the serial number has not been set, no a_serial is returned.

Example: VER
VER
VER MultidropCombi 1.00.20
VER END 0

Service commands

The Multidrop Combi service commands are presented below.

BOOT – enter bootstrap load mode

This is another exception to the 3-character command length. The microcontroller is put to a simulated bootstrap load mode. The bootstrap load mode is described in the User's Manual for the C167 Derivatives. Refer to <http://www.infineon.com>.

Before entering the bootstrap load mode, the BOOT END 0 response is sent. After receiving the response, the PC can start the boot process through the serial port. However, a small delay is recommended because the boot mode is not entered until the PC has acknowledged the successful reception of the END response, including the <CR> and <LF> (USB protocol).

If this command is used for starting the download of a new firmware and the download fails for any reason, the only way to recover is to switch off instrument power, insert the bootstrap jumper on the controller board, switch instrument power on and perform the download through the serial port the same way as after sending the BOOT command.

Command: BOOT
Response: BOOT
BOOT END ui_status

The BOOT command is executed even if the instrument has an unacknowledged error.

GET – report user defined parameters

This command reports all or selected user-defined parameters. The user-defined parameters are name and value combinations stored to the nonvolatile memory with the PUT command.

Command: GET [a_name1 ... a_name_n]

a_name_x Name given for the user-defined parameter. This must match the name used when the parameter was stored with the PUT command. The name may include the '=' character which is used for separating the name from the value string. If no names are given, all name/value pairs are reported.

Response: GET
GET a_name₁=a_value₁
:
GET a_name_n=a_value_n
GET END ui_status

a_name_x Name of the parameter.

a_value_x The ASCII string value of the parameter.

If there are no user-defined parameters stored, no data responses are returned. Also, if the parameter named in the command is not found, there will be no response line for that parameter.

Example: *GET*
:
GET
GET numbers=0,1,2,3,4,5,6,7,8,9
GET user1=Hannes&Okkelstrom
GET END 0

HEX – send a hex file record

This command offers an alternative method besides the BOOT command for updating the firmware.

Command: HEX a_record

a_record An Intel HEX file record. Each character in the record must be one of '0...9', 'a...f' or 'A...F' except the first one, which must be ':'.
:

Response: HEX END ui_status

The PC should send the whole HEX file one record at a time. The first record sent must be a segment address record or an extended linear address record with the address 0x000000. For example,

HEX :0200000020000FC.

The last record must be the end-of-file record:

HEX :00000001FF

Between the first and last record any number of address and data records may be sent. Even other than HEX commands may be sent, as long as the command does not cause a write to the nonvolatile parameters. All the data in the data records must fall within the firmware code memory.

After a valid end-of-file record, the programming of the code memory starts as soon as the END response is sent. When the programming is complete, the instrument is reset by software, and a normal startup sequence is executed.

If the programming fails for any reason, the only way to recover is to switch off instrument power, insert the bootstrap jumper on the controller board, switch instrument power on, and perform the download through the serial port the same way as after sending the BOOT command.

LOG – send error log

Whenever the firmware detects an error, the error information is stored in the nonvolatile memory. Some fatal errors cause the firmware to halt, but even then the error information is first stored unless the error was in writing to the nonvolatile memory. With this command, the error log can be read by the PC.

The time of the error is recorded, but if the calendar clock of the instrument is not running, it will always be reported as 01.01.2000

00:00. It is recommended that every time the PC connects to the instrument it starts the calendar clock by using the TIM command.

Up to 100 errors are stored. After the 100th error, each new error overwrites the oldest stored error.

The errors are reported the oldest one first and the latest last, no matter what the date and time in the clock calendar was at the time of error.

Command: LOG

Response: LOG

LOG ui_errcode dd.mm.yyyy HH:MM [a_module ui_line]

:

LOG END ui_status

ui_errcode The error code. See command EAK for possible error codes.

dd Day of month of the error. This is always printed with two digits, 01...31.

mm Month of error. This is always printed with two digits, 01...12.

yyyy Year of error, 2000...9999.

HH Hour of error. A 24-hour format is used and the hour is always printed with two digits, 00...23.

MM Minutes of error. This is always printed with two digits, 00...59.

a_module This is optional and is only printed for error code 1. It is the name of the firmware source code module where the error occurred.

ui_line This is optional and is only printed for error code 1. It is the line number of source code module of firmware where the error occurred.

Example: LOG

:

LOG

LOG 5 23.04.2002 14:35

:

LOG 1 26.04.2002 08:13 MDX.C 2246

LOG END 0

In this example the oldest error in the log was 'Plate X position error', which occurred on 23. April 2002 at 14:35. The latest error is 'Firmware error in module MDX.C on line 2246'. This occurred on 26. April 2002 at 8:13. The latter kind of errors should not appear.

In addition to the error codes listed in the EAK command description, the following error codes may appear in the error log (Table 3–3).

Table 3–3. Error codes

Code	Explanation
1007	This value stands for a collection of possible RTX error causes.
1008	Invalid RTX call parameter value.
1009	Invalid RTX call handle.
1010	A RTX call requested a handle, but none is available.
1011	There was an inconsistency in system data used by RTX.
1012	RTX could not allocate memory.
1013	Interval was already expired when a wait for interval was started (RTX).
1014	Attempt to create an object which the configuration does not support (RTX).
1015	RTX function was not executed.
1016	An object a task was waiting at was deleted (RTX).
1017	Function not allowed for this mailbox type (RTX).
1256	Unexpected NMI interrupt.
1257	Undefined opcode.
1258	Protected instruction fault.
1259	Illegal word operand access.
1260	Illegal instruction access.
1261	Illegal external bus access.
1262	Stack overflow caused by RTX task.
1263	Stack overflow exceeding stack reserve.
1264	Second stack overflow trap in sequence.
1265	Stack underflow caused by RTX task.
1266	Stack underflow exceeding valid stack area.
1267	Unexpected interrupt, for which there is no service.

These errors are generated either by the used Real Time Executive (RTX166), or by the C167 microcontroller. RTX generated errors cause the firmware to halt, and cycling the power down is required to restart the instrument. C167 generated errors usually indicate a hardware problem. These errors also cause the firmware to halt, but due to the hardware nature of the problem, the processor may actually run random code instead of halting. For more information about these errors, refer to the User's Manuals for the RTX166 and for the C167 (<http://www.keil.com>).

If the instrument stops operating unexpectedly and/or any of these errors are listed in the error log, contact service.

MOP – move pump

This command turns the pump rotor to the given angle from the home position.

Command: MOP ui_angle

ui_angle

The angle to turn the rotor to, 0...89 degrees. Angle 0 is the rotor home position and the rotor is turned counterclockwise when viewed from the front of the instrument.

Response: MOP
MOP END ui_status

One pump motor step corresponds to a rotation of 0.9 degrees of the pump rotor. The given angle is rounded to the nearest step position.

Example: *MOP 56*
MOP
MOP END 0

MOX – move plate carrier

This command moves the plate carrier to the given position.

Command: MOX si_pos

si_pos

Target position in 1/100 mm. Range – 3500...11850. Position 0 is where the left end of a plate installed into the plate carrier is under the dispensing tips. The position increases when the plate moves left.

Response: MOX
MOX END ui_status

Example: *MOX 123*
MOX
MOX END 0

MOY – move Y

This command moves the Y to the given position.

Command: MOY si_pos

ui_pos

Target position in 1/100 mm. Range 700...1675. Position 0 is where the front edge of a plate installed into the plate carrier is under the dispensing tip closest to the user. The position increases when the tips move away from the user.

Response: MOY
MOY END ui_status

Example: *MOY 123*
MOY
MOY END 0

MOZ – move Z This command moves the dispensing tips to the given height.

Command: MOZ ui_height

ui_height Tips' height from the bottom of a plate inserted into the plate carrier. This is in 1/100 mm and the allowed range is 500...5500.

Response: MOZ
MOZ END ui_status

Example: *MOZ 123*
MOZ
MOZ END 0

PAR – set nonvolatile parameters This command is used to set nonvolatile parameters of the instrument. Related parameters are grouped under the same group code.

Command: PAR ui_group values

ui_group The parameter group to set.

values List of values to set for the parameters.

Response: PAR
PAR END ui_status

Nonvolatile parameters which can be set by the PAR command:

Group 0 Set instrument serial number.

Command: PAR 0 a_serial_number

a_serial_number Instrument serial number as an ASCII string. Allowed characters: '!'...'~'. '&' translates into white space.



Note The serial number can only be set once. ▲

Group 1 Set instrument name.

Command: PAR 1 a_name

a_name Instruments name as an ASCII string. Allowed characters: '!'...'~'. '&' translates into white space. The instrument responds with this name to the VER command. The default name is 'MuptidropCombi'.

Group 2 Set X/Y/Z home offsets. This command is used to adjust the X, Y and Z positions relative to the home sensor position, that is, to correct the sensor position error.

Command: PAR 2 si_x_offset si_y_offset si_z_offset

si_x_offset	X position adjustment in 1/100 mm, -100...100.
si_y_offset	Y position adjustment in 1/100 mm, -100...100.
si_Z_offset	Z position adjustment in 1/100 mm, -100...100.

Note that the offsets are stored in steps. As the step resolution is not 1/100 mm, the offsets reported in REP 5 may slightly differ from the offsets set with the PAR 2 command.

PPA – set remote pump move parameters

This command sets pump move parameters for the remote operations.

Command: PPA ui_acceleration ui_deceleration [ul_post_delay]

ui_acceleration	The acceleration used for the pump motor, 1...100. This is in % of the difference of the maximum and the minimum pump acceleration, and is added to the minimum acceleration. Value 0 is also accepted and does not change the current acceleration.
ui_deceleration	The deceleration used for the pump motor, 1...100. This is in % of the difference of the maximum and the minimum pump acceleration, and is added to the minimum acceleration. Value 0 is also accepted and does not change the current deceleration.
ul_post_delay	This parameter defines the wait time before the plate can be moved after dispensing. It is in ms. The delay applies to the current run protocol, that is, the setting is not global.

Response: PPA
PPA END ui_status

PUT – save user-defined parameter to nonvolatile memory

This command is used for storing any user-defined parameters to the nonvolatile memory of the instrument. The stored parameters can be queried with the GET command. This command is also used for deleting user-defined parameters from the nonvolatile memory.

Command: PUT a_name=[a_value]

a_name	The name the user gives for the parameter string. Allowed characters: '!'...'~', except '=' which separates a_name from a_value. '&' translates into white space.
--------	---

The name must be followed by the '=' character.
a_value Parameter value. This is an ASCII string encoded any way the user of the command wishes. Allowed characters: '!'...'~'. '&' translates into white space. If this parameter is not given, the parameter string named a_name is deleted from the nonvolatile memory.

Response: PUT
PUT END ui_status

The amount of nonvolatile memory allocated for user-defined parameters is 1024 bytes. For each parameter entry 2 bytes is used for parameters management. Also, for technical reasons, the size of each entry is rounded up to the nearest even number of bytes. If there is not enough memory, an error is reported.

If the named parameter already exists, it is first deleted before writing the new value.

Since the number of times the nonvolatile memory can be reprogrammed is limited, it is not recommended to use this command for storing information, which frequently changes.

Examples: *PUT numbers=0,1,2,3,4,5,6,7,8,9*
PUT
PUT END 0
PUT user1=Hannes&Okkelstrom
PUT
PUT END 0

REA – read a byte or word value

This command reads a byte or word value from the given memory address of the microcontroller of the instrument. This command is intended for diagnostics use only. Incorrect use may crash the internal software or cause undefined behavior.

Command: REA a_size a_address

a_size Data size specifier, a single letter.

B byte
W word

a_address Address to read in hexadecimal notation. Hex digits '0..9', 'a..f' and 'A..F' are allowed.

Response: REA
REA ui_decvalue a_hexvalue
REA END ui_status

ui_decvalue Value read in decimal notation.

a_hexvalue Value read in hexadecimal notation. For a byte read two and for a word read four hexadecimal digits are always printed.

The address must be in the range 0...FFFFFF. A word read from an odd address is illegal.

Example: *REA WFF1E*
REA
REA 65535 FFFF
REA END 0

The above command reads a word from the hex address FF1E. The ONES register of the microcontroller is at that address and thus the shown response is always returned.

REP – report

This command can be used to acquire a variety of instrument parameters and settings. They are grouped into sections of related parameters or settings. Each section is identified with a section code, which is used in the responses and can also be used in the command.

Command: REP [ui_section1 ... ui_sectionn]

ui_section_x Which section(s) to report, 0 to the last section documented for this command. This is optional. If no sections are requested, all sections are reported.

Response: REP
REP ui_section values
:
REP END ui_status

ui_section The section reported in this response line.
values List of values reported in this section. See the description of values for each section.

If there is nothing to report for any section, the report line is not returned. For instance, if the instrument serial number was requested but it is not set, no REP 0 line is returned.

List of values reported for each section:

Section 0 Instrument serial number.
REP 0 a_serial_number

a_serial_number Instrument serial number as an ASCII string.

Section 1 Instrument name.
REP 1 a_name

a_name Name of the instrument as an ASCII string. The instrument responds with this name to the VER command. The default name is 'MultidropCombi'.

Section 2 Cassette definitions.
REP 2 ui_cassette_id a_name ui_min_vol ui_max_vol ui_step_vol
ui_min_speed ui_max_speed ui_rotor_offset

ui_cassette_id Cassette id. (0...3)
0 = Standard cassette (factory settings)

	1 = Small cassette (factory settings)
	2 = User-defined place
	3 = User-defined place
a_name	Cassette name. Maximum length of the name is 20 characters.
ui_min_vol	Minimum volume in 1/10 µl.
ui_max_vol	Maximum volume in 1/10 µl.
ui_step_vol	Smallest volume increment in 1/10 µl.
ui_min_speed	Minimum speed 1...100%.
ui_max_speed	Maximum speed 1...100%.
ui_rotor_offset	Rotor offset, -200...200 (-20...20 degrees).

Section 3

Plate definitions.

REP 3 ui_plate_type a_name ui_first_column_x ui_last_column_x
ui_first_row_y ui_last_row_y ui_columns ui_rows ui_plate_height
ui_max_well_volume

ui_plate_type	Plate type 0...30 The following types are factory defined: 96 standard (15mm) 0 96 low DW (22mm) 1 96 DW (44mm) 2 384 low vol. (7.5mm) 3 384 low prof. (10mm) 4 384 standard (15mm) 5 384 DW (22mm) 6 384 DW (44mm) 7 1536 low vol. (5mm) 8 1536 std (10.5mm) 9 Types 10...29 are user defined. Plate type 30 is the plate defined with the PLA command and used by the DIS command.
a_name	Plate name. Maximum length of the name is 20 characters.
ui_first_column_x	Distance from the left edge of the plate to the center point of the wells in the first column, in 1/100 mm.
ui_last_column_x	Distance from the left edge of the plate to the center point of the wells in the last column, in 1/100 mm.
ui_first_row_y	Distance from the front edge of the plate to the center point of the wells in the first row, in 1/100 mm.
ui_last_row_y	Distance from the front edge of the plate to the center point of the wells in the last row, in 1/100 mm.
ui_columns	Number of well columns on the plate, 1...48.
ui_rows	Number of well rows on the plate, 8, 16 or 32.
ui_plate_height	Height of the plate 0...5000.

ui_max_well_volume Maximum volume to dispense into a well in 1/10 µl, 5...25000.

Section 4

Protocol definitions.

REP 4 a_protocol_name ui_plate_type ui_cassette_type ui_volume1
ui_volume2...ui_volume48 ui_selector_channel ui_speed
si_dispense_x_offset si_dispense_y_offset ui_dispensing_height
ul_prime_volume ui_order ul_post_delay ui_attribute

a_protocol_name Name of a protocol stored in the nonvolatile memory of the instrument. Maximum length of the name is 20 characters.

ui_plate_type Plate type 0...30
The following types are set as factory settings (see Appendix A: "Plate Definitions"):

96 standard (15mm)	0
96 low DW (22mm)	1
96 DW (44mm)	2
384 low vol. (7.5mm)	3
384 low prof. (10mm)	4
384 standard (15mm)	5
384 DW (22mm)	6
384 DW (44mm)	7
1536 low vol. (5mm)	8
1536 std (10.5mm)	9

Plate type 30 is the plate defined with the PLA command and used by the DIS command.

ui_cassette_type 0 = standard
1 = small
2 = user defined
3 = user defined

ui_volume1 0 = do not dispense into column 1
5...25000 dispensing volume in 1/10 µl

ui_volume2 0 = do not dispense to column 2
:
5...25000 dispensing volume in 1/10 µl

:
ui_volume48 0 = do not dispense into column 48
5...25000 dispensing volume in 1/10 µl

ui_selector_channel Selector channel the protocol uses, 0...6.
Channel 0 means not used/do not care.

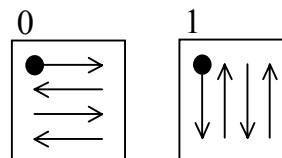
ui_speed The top speed used for the pump motor during dispensing, 1...100. This is in % of the difference of the maximum and the minimum speeds of the cassette, and is added to the minimum speed of the cassette.

si_dispense_x_offset The range is -300...300 [1/100 mm].

si_dispense_y_offset The range is -300...300 [1/100 mm].

ui_dispensing_height Height of the dispensing 0...5000.

ul_prime_volume	This is the predispensing volume in 1/10 µl, 20...100000.
ui_order	Value 0 dispenses row wise and 1 column wise. The figures below illustrate the corresponding filling sequences.



ul_post_delay	For an 8-row plate the selection of the dispensing order makes no difference. This parameter defines wait time before the track is moved to the next dispensing position. This is in ms.
ui_attribute	A combination of the following property flags: 1 = Read-only protocol 2 = Start-up protocol 4 = Default protocol If no flags are set (ui_attribute = 0), the protocol is a normal protocol. See command WPR.

Section 5 Stepper home offsets.

REP 5 si_x_offset si_y_offset si_z_offset ui_x_ref_index ui_y_ref_index ui_z_ref_index si_plate_out_offset

si_x_offset	X home position offset in 1/100 mm.
si_y_offset	Y home position offset in 1/100 mm.
si_z_offset	Z home position offset in 1/100 mm.
ui_x_ref_index	X stepper step pattern index at home sensor. Not currently used, always 0.
ui_y_ref_index	Y stepper step pattern index at home sensor. Not currently used, always 0.
ui_z_ref_index	Z stepper step pattern index at home sensor. Not currently used, always 0.
si_plate_out_offset	Plate out position adjustment in 1/100 mm. Not currently used, always 0.

Note that since the offsets are stored in steps and the step resolution is not 1/100 mm, the values reported in REP 5 may slightly differ from the offsets set with the PAR 2 command.

Section 6 Current stepper positions.

REP 6 si_pump_pos si_x_pos si_y_pos si_z_pos si_selector_pos

si_pump_pos	Current pump stepper position in 1/2 steps from the previous 1/4 revolution position (in dispensing direction), 0...99.
-------------	---

si_x_pos	Current X stepper position in 1/8 steps from the home sensor.
si_y_pos	Current Y stepper position in 1/8 steps from the home sensor.
si_z_pos	Current Z stepper position in 1/2 steps from the home sensor.
si_selector_pos	Current selector channel, 1...6. If the position is not known, a question mark (?) is reported for that stepper.

Section 7 Report startup status.
REP 7 ui_item ul_value

ui_item	Report item number, 0...6. Each item is reported on a separate response line. 0 = Number of 10 ms ticks elapsed since startup. 1 = Pump startup status. 2 = X stepper startup status. 3 = Y stepper startup status. 4 = Z stepper startup status. 5 = Selector startup status. 6 = Nonvolatile parameters startup status.
ul_value	For item 0 the value is the current count in the 10 ms system tick counter. For other items the value is one of 0, 1 or 2: 0 Not tested. 1 Startup test passed. 2 Startup test failed.

Section 8 Dynamic memory allocation usage.
REP 8 ul_peak_usage ul_current_usage

ul_peak_usage	Maximum number of bytes of dynamic memory used any time after startup.
ul_current_usage	The number of bytes of dynamic memory currently allocated.

RFS – restore factory settings

This command deletes all current protocols, plates and cassettes from the nonvolatile memory and then writes the factory-defined protocols, plates and cassettes to the nonvolatile memory.

Command: RFS
Response: RFS
RFS END ui_status

SRE – status report

Switch status reporting on or off. When on, SRE responses are sent to the computer while executing commands. These give additional

information of the current instrument status. After startup status reporting is off.

Command: SRE ui_mode

uc_mode 0 = reporting OFF
 1 = reporting ON.

Response: SRE
SRE END ui_status

When status reporting is on, the following status responses are sent:

SRE 0

The instrument has entered idle state and is waiting for a new command.

SRE 1

Working. This is always reported when the execution of a new command starts, if the previous report was not SRE 1. For most of the commands there are no dedicated status codes for their execution and this is the only status reported.

SRE 2

Dispensing. Reported when dispensing starts.

SRE 3

Priming. Reported when priming the hoses starts.

SRE 4

Emptying. Reported when emptying the hoses starts.

SRE 5 ul_time

Shaking. This is reported when executing the SHA command.

ul_time The remaining shaking time in 1/100 s. If continuous shaking, ul_time is 0.

SRE 6

Aborting. Reported when an Abort command is received.

SRE 7 ui_column ui_row

Completed dispensing 8 wells of a column. When status reporting is on, this response is always sent to the computer during execution of the DIS command when dispensing into 8 wells of a column ends.

ui_column Column number dispensed, 1...48.

ui_row Lowest number row dispensed, 1...4. Since the instrument is an eight-channel dispenser, seven other rows were also dispensed into. For a 96-well plate, ui_row is always 1 and all rows 1...8 are dispensed simultaneously. For a 384-well plate, ui_row is 1 or 2 and all even numbered rows or all odd numbered rows are dispensed.

For a 1536-well plate, ui_row is 1...4 and every fourth row is dispensed, starting from the row in the response.

WCA – write cassette

This command sets the parameters of the given cassette type and saves them to the nonvolatile memory. Cassettes 0 and 1 are factory defined and error 45 is returned if ui_cassette_id 0 or 1 is used in the command. Cassettes 2 and 3 are free to use. The REP 2 command returns cassette definitions from the device.

The cassette index is set as free when the WCA command with index and name (no other parameters) is sent.

Command: WCA ui_cassette_id a_name [ui_min_vol ui_max_vol ui_step_vol ui_min_speed ui_max_speed si_rotor_offset]

ui_cassette_id	Cassette id, 2 or 3.
a_name	Cassette name. Maximum length of the name is 20 characters. Allowed characters: 'A'...'Z', 'a'...'z', '0'...'9', '-', '_' and '&' which translates into white space.
ui_min_vol	Minimum volume in 1/10 µl, 1...25000. This must be a multiple of ui_step_vol and may not be greater than ui_max_vol.
ui_max_vol	Maximum volume in 1/10 µl, 1...25000. This must be a multiple of ui_step_vol.
ui_step_vol	Step volume in 1/10 µl, 1...250. This corresponds to 1/8 pump revolution.
ui_min_speed	Minimum speed 1...100%.
ui_max_speed	Maximum speed 1...100%.
si_rotor_offset	Rotor offset, -200..200 (-20..20 degrees).

The speed limits are used for limiting the speed range of the cassette to a portion of the total pump speed range. For example, a minimum cassette speed of 30% would set the minimum dispensing speed of the cassette to minimum pump speed + 30% of the pump speed range.

Response: WCA
WCA END ui_status

Example: Add new cassette definition

```
WCA 2 My&Cassette 50 25000 50 85 0 100
WCA
WCA END 0
```

Release place 2:

```
WCA 2 My&Cassette
```

WRI – write a byte or word value

This command writes a byte or word value to the given memory address of the microcontroller of the instrument. This command is intended for diagnostics use only. By using this command you may crash the internal software, cause undefined behavior and even make the instrument unusable. Do not use this command unless you know exactly what you are doing.

Command: WRI a_size a_address ui_value

a_size	Data size specifier, a single character.
B	byte
W	word
a_address	Address to write to in hexadecimal notation. Hex digits '0...9', 'a...f' and 'A...F' are allowed.
ui_value	Value to write in hexadecimal notation. Hex digits '0...9' and 'A...F' are allowed.

Response: WRI
WRI END ui_status

The allowed ranges for the address are: E000 – FFFF, 800000 – 8FFFFFF, 910000 – 917FFF, A00000 – BFFFFFF. The value for a byte write must be 0...FF and for a word write 0...FFFF. Word writes to an odd address are illegal. Writing to address range 910000 – 917FFF may corrupt the nonvolatile parameters.

Example: *WRI B B20000 A5*
WRI
WRI END 0

This command writes the first pixel of the display.



Caution Writing any value to address 910000 deletes the nonvolatile parameters. ▲

WPL – write plate

This command writes the given plate definition to the Flash.

The REP 3 command returns plate definitions from the device.

Command: WPL ui_plate_type a_name [ui_first_column_x ui_last_column_x ui_first_row_y ui_last_row_y ui_columns ui_rows ui_plate_height ui_max_well_volume]

ui_plate_type	Plate type 10...29. Plate types 0...9 are factory defined, and error 45 is returned if ui_plate_type 0...9 is used.
a_name	Plate name. Maximum length of the name is 20 characters. Allowed characters: 'A'...'Z', 'a'...'z', '0'...'9', '-', '_' and '&' which translates into white space.
ui_first_column_x	Distance from the left edge of the plate to the center point of the wells in the first column. See

	command PLA for limits.
ui_last_column_x	Distance from the left edge of the plate to the center point of the wells in the last column. See command PLA for limits.
ui_first_row_y	Distance from the front edge of the plate to the center point of the wells in the first row. See command PLA for limits.
ui_last_row_y	Distance from the front edge of the plate to the center point of the wells in the last row. See command PLA for limits.
ui_columns	Number of well columns on the plate, 1...48.
ui_rows	Number of well rows on the plate, 8, 16 or 32.
ui_plate_height	Height of the plate 0...5000.
ui_max_well_volume	Maximum volume to dispense into a well, 1...25000 (0.1...2500 µl). Note that the maximum volume during dispensing is the smaller of this volume and the maximum dispensing volume of the used cassette.

Response: WPL
WPL END ui_status

The plate type is removed from the Flash when the WPL command is given with only the plate type and name (no other parameters). For example, WPL 23 My&Plate.

All coordinates and distances in the command are in 1/100 mm.

Error 44 is returned if the plate type defined in the command is the one currently used by the start-up protocol or the protocol currently in the user interface.

The distance between the first row and the last row must be 63 mm $\pm$ 1 mm for 8 rows, 67.5 mm $\pm$ 0.5 mm for 16 rows, and 69.75 mm $\pm$ 0.1 mm for 32 rows.

The origin of the coordinates is at the front left corner of a plate inserted into the plate carrier of the instrument.

Figure 3–2 shows how the distance arguments of the command are defined.

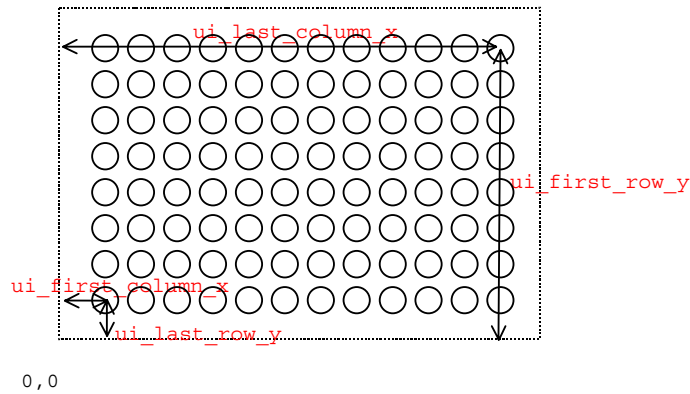


Figure 3–2. Distance arguments of the command

**WPR – write
protocol**

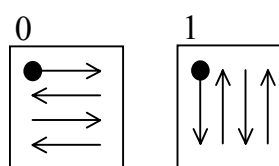
This command writes the protocol definition for a given plate type to the nonvolatile memory.

The REP 4 command returns protocol definitions from the device.

Command: WPR a_protocol_name ui_plate_type ui_cassette_type ui_volume1 ui_volume2...ui_volume48 ui_channel ui_speed si_dispense_x_offset si_dispense_y_offset ui_dispensing_height ul_prime_volume ui_order ul_post_delay ui_attribute

a_protocol_name	Name of a protocol. Maximum length of the name is 20 characters. Allowed characters: 'A'...'Z', 'a'...'z', '0'...'9', '-', '_' and '&' which translates into white space. If ui_attribute is Default protocol, a_protocol_name must be Default&protocol .																				
ui_plate_type	Plate type the protocol uses, 0...29. The following types are factory defined (see Appendix A: "Plate Definitions"): <table><tr><td>96 standard (15mm)</td><td>0</td></tr><tr><td>96 low DW (22mm)</td><td>1</td></tr><tr><td>96 DW (44mm)</td><td>2</td></tr><tr><td>384 low vol. (7.5mm)</td><td>3</td></tr><tr><td>384 low prof. (10mm)</td><td>4</td></tr><tr><td>384 standard (15mm)</td><td>5</td></tr><tr><td>384 DW (22mm)</td><td>6</td></tr><tr><td>384 DW (44mm)</td><td>7</td></tr><tr><td>1536 low vol. (5mm)</td><td>8</td></tr><tr><td>1536 std (10.5mm)</td><td>9</td></tr></table>	96 standard (15mm)	0	96 low DW (22mm)	1	96 DW (44mm)	2	384 low vol. (7.5mm)	3	384 low prof. (10mm)	4	384 standard (15mm)	5	384 DW (22mm)	6	384 DW (44mm)	7	1536 low vol. (5mm)	8	1536 std (10.5mm)	9
96 standard (15mm)	0																				
96 low DW (22mm)	1																				
96 DW (44mm)	2																				
384 low vol. (7.5mm)	3																				
384 low prof. (10mm)	4																				
384 standard (15mm)	5																				
384 DW (22mm)	6																				
384 DW (44mm)	7																				
1536 low vol. (5mm)	8																				
1536 std (10.5mm)	9																				
ui_cassette_type	Cassette type the protocol uses: 0 = Standard tube cassette (factory defined) 1 = Small tube cassette (factory defined) 2 = user defined 3 = user defined																				

ui_volume1 : :	Dispensing volume for column 1 in 1/10 µl. 0 = do not dispense into column 1. If not 0, the maximum is 25000 (2.5 ml). The minimum depends on the cassette type: 50 (5 µl) for the Standard tube cassette and 5 (0.5 µl) for the Small tube cassette.
ui_volume48	Dispensing volume for column 48 in 1/10 µl. 0 = do not dispense into column 48. If not 0, the maximum is 25000 (2.5 ml). The minimum depends on the cassette type: 50 (5 µl) for the Standard tube cassette and 5 (0.5 µl) for the Small tube cassette.
ui_channel ui_speed	Selector channel, 1...6, or 0 if selector not used. The top speed used for the pump motor during dispensing, 1...100. This is in % of the difference of the maximum and the minimum cassette speeds, and is added to the minimum cassette speed.
si_dispense_x_offset si_dispense_y_offset	The range is -300...300 1/100 mm. The range is -300...300 1/100 mm.
ui_dispensing_height ul_prime_volume	Height of the dispensing 500...5500. This is the predispensing volume in 1/10 µl. The upper limit is 100000, the lower equals the minimum dispensing volume of the cassette. Note that the value is checked against the cassette limits when the protocol is started, and may result in an error if it is not within the limits.
ui_order	Value 0 dispenses row wise and 1 column wise. The figures below illustrate the corresponding filling sequences.



ul_post_delay	For an 8-row plate the selection of the dispensing order makes no difference. This parameter defines wait time in ms, before the track is moved to the next dispensing position. The default value is 0.
ui_attribute	Protocol attributes. This is a combination of zero or more property flags: 1 Read-only protocol, cannot be deleted or modified. 2 Start-up protocol, loaded to UI when instrument starts up. 4 Default protocol (for a plate type), loaded when a plate type is selected

from UI.
8 Delete the protocol if it exists
(matching name and plate type).
Any combination of flags 1, 2 and 4 can be
used, for example, attribute 6 would be Default
protocol and Start-up protocol. If no flags are
set (ui_attribute = 0), the protocol is a normal
protocol.

Response: WPR

WPR END ui_status

Example: *WPR MyProtocolName 13 0 100 100 100 100 100 100 0 100 100
100 100 100
0 0 0 78 -12 -45 1345 5000 0 0 0*

Error 44 is returned in the following situations:

- The command would change or delete the protocol currently loaded in the user interface.
- The start-up protocol attribute is set, the default protocol attribute is not, and there is no default protocol for the plate the protocol uses.
- The start-up protocol attribute is not set and the protocol would replace the current start-up protocol.
- The command would reset the default protocol attribute of an existing protocol and the start-up protocol uses the same plate as the existing protocol.

These rules are due to the following facts:

- There must always be a start-up protocol.
- There must always be a default protocol for the plate type the start-up protocol uses.
- The protocol currently in the user interface must always be valid.

Error 45 is returned if the command would modify or delete a read-only protocol.

Error 3 is returned if the default protocol attribute is set and the name of the protocol is not “**Default&protocol**”. Only protocols using cassette type 0 or 1 are shown in the user interface. Protocols using a certain plate do not show up in the user interface until a default protocol has been defined for the plate.

If the startup attribute is set, the cassette type must be 0 or 1.

The ui_plate_type and ui_cassette_type arguments must select an existing plate and cassette. If you want to create a protocol using a new plate and/or cassette type, define the plate and/or cassette first with the WPL and WCA commands, respectively.

Truth table for changing the attributes of an existing protocol is shown in Table 3–4.

Table 3–4. Truth table for changing the attributes of an existing protocol

From	To								
	0	1	2	3	4	5	6	7	8
0	OK	OK	OK ¹	OK ¹	OK ⁰	OK ⁰	OK ⁰	OK ⁰	OK
1	error 45	OK ²	error 45	OK ³	error 45	OK ^{0,2}	error 45	OK ^{0,2}	error 45
2	error 44	error 44	OK	OK	0.6	0.6	0.6	0.6	error 44
3	error 45	error 45	error 45	OK ²	0.6	0.6	0.6	0.6	error 45
4	OK ⁴	OK ⁴	error 44	error 44	OK ⁰	OK ⁰	OK ⁰	OK ⁰	OK ⁴
5	error 45	OK ⁵	error 45	OK ⁵	error 45	OK ^{0,2}	error 45	OK ^{0,2}	error 45
6	error 44	error 44	error 44	error 44	error 44	error 44	OK ⁰	OK ⁰	error 44
7	error 45	error 44	error 45	error 44	error 45	error 44	error 45	OK ^{0,2}	error 45

- 0 If the default protocol attribute is set, the name of the protocol must be 'Default protocol'.
- 1 Possible only if a default protocol for the plate exists, else error 44 is returned.
- 2 Possible only if the protocols are identical, else error 45 is returned.
- 3 Possible only if the protocols are identical and a default protocol for the plate exists, else error 44 or 45 is returned.
- 4 Possible only if the start-up protocol does not use the same plate, else error 44 is returned.
- 5 Possible only if the protocols are identical and the start-up protocol does not use the same plate, else error 44 or 45 is returned.
- 6 It is not possible to have a start-up protocol named 'Default protocol' unless the default protocol attribute is also set. Therefore, the situation of adding the default protocol attribute to a protocol having the start-up protocol attribute is not possible.



Note There is space for 100 protocols in the nonvolatile memory. These include the factory-defined protocols. ▲

Chapter 4

Emulated Command Set

The new Multidrop Combi emulates the commands and responses of two old Multidrop versions: Multidrop 384 V1.7 and Multidrop Micro V1.1. The emulation mode is selected from the user interface. In emulation mode any command sets the instrument in the computer control mode unless the instrument is busy because it is being operated from the user interface. In this case the instrument does not respond to the command.

The user can exit the computer control mode by pressing the **STOP** key. If this is done while a command from the computer is being executed, there will be no response to that command.

Commands

Commands of the emulated Multidrop models have two to three fields as described in the following syntax. The first letter identifies the command. Then a numeric argument field is optional for some and required for other commands. The numeric field immediately follows the uppercase letter without any separating characters. The last character of a command is a linefeed character **<LF>** or a carriage return character **<CR>**. Commands may also be terminated with **<CR><LF>** or **<LF><CR>**, since the instrument ignores empty commands. Unless noted, the commands are similar for both emulation modes.

In the following list of commands, the command arguments in square brackets [] are optional. If the optional argument is left out, the default value is used. See Table 4–5.

- | | |
|-----------------------|--|
| D<LF> | Dispenses the volume set by the “Vv” command to the whole plate. |
| E[v]<LF> | Empties the pump. |
| G<LF> | Dispenses the plate using the protocol currently in the user interface. This command has the same effect as pressing the START button when not in remote control. |
| M[c]<LF> | Dispenses the given number of columns starting from the current column. If the column count is not given, one column is dispensed. The dispensing volume is set by the “Vv” command. After the command is completed, the dispensing tips remain at the last column dispensed. If this command is received when the plate is in the home position, dispensing starts from column 1. ER3 is reported if more columns are requested than left over from the current column to the last column of the plate. |

N<LF>	Reports the emulated software version.
O<LF>	Drives the plate out.
P[v]<LF>	Primes the given volume.
Q<LF>	Resets the instrument. There is no response to this command.
R[r]	Sets the rotor speed. It is possible to change the rotor speed (dispensing speed) through this command. This command is only available in the Multidrop Micro emulation.
S[c]<LF>	Drives the requested column under the dispensing tips. Valid columns are 1 to 12 for a 96-well plate and 1 to 24 for a 384-well plate. If the column is not given, the instrument steps forward one column, or to plate out from the last column, or to the first column from plate out. c = 1 – 12 96-well plate c = 1 – 24 384-well plate
Tt<LF>	Sets the plate type assumed for all other commands except the “G” command. t = 0 96-well plate t = 1 384-well plate
V<LF>	Reports the internal software version of the instrument
VER<LF>	just like the “N” command.
Vv<LF>	Sets the volume for dispensing.
Zs<LF>	Shakes the plate for the requested time.



Note After starting or resetting the instrument all values are returned to default. ▲

Default values and limits in the two emulation modes are presented in Table 4–5.

Table 4–5. Default values and limits in the two emulation modes

Command	Multidrop 384 emulation	Multidrop Micro emulation
Default dispensing volume (D or M command before Vv command)	20 µl	5 µl
Predispensing volume (volume predispensed at the beginning of D command)	10 µl	2 µl
E command default volume	880 µl	150 µl
E command limits	Volume not allowed	1-999 µl
P command default	200 µl	120 µl
P command limits	5-1000 µl in 5 µl increments	1-999 µl
R command default	—	6
R command limits	—	0..6
Default plate type before T command	Standard 96-well plate	Standard 96-well plate
Vv command limits	5-1000 µl in 5 µl increments	1-50 µl
Z command limits	1-60	1-60

Responses and error codes

The instrument sends a response to each command after the command has been executed. The only exception is the “Q” command, for which there is no response.

The response to the “N”, “V” or “VER” command is as follows:

Multidrop 384 V1.7 emulation:

Mdrop384 V1.7<CR><LF>

Multidrop Micro V1.1 emulation:

MDMicro V1.1<CR><LF>

The response to other commands, if the command is executed successfully, is as follows:

OK<CR><LF>

If an error is detected, the response is one of the following:

- ER3<CR><LF>** Unrecognized command or invalid command argument.
- ER4<CR><LF>** The pump is not primed.
- ER5<CR><LF>** The priming vessel is not inserted into its slot.
- ER6<CR><LF>** Hardware error.
- ER7<CR><LF>** The pump has lost steps. The next priming, manual or in conjunction with the “D” command, will readjust the rotor home position. This error is only reported in the Multidrop Micro emulation. The Multidrop 384 emulation returns ER6 instead.

Default protocols

The instrument emulates serial interface commands of old Multidrop Micro and Multidrop 384 instruments. Both of these emulation modes have own default protocols for 96- and 384-well plates.

Multidrop Micro protocol settings for a 96-well plate:

- Plate is 96 standard (15mm).
- Small (tube) cassette.
- Volume is 5 µl into all columns.
- Speed is 100%.
- X and Y track offsets are 0.
- Dispensing height is 16 mm.
- Predispense volume is 2 µl.
- Dispensing direction is 'by row'.

Multidrop Micro protocol settings for a 384-well plate:

- Plate is 384 standard (15mm).
- Small (tube) cassette.
- Volume is 5 µl into all columns.
- Speed is 100%.
- X and Y track offsets are 0.
- Dispensing height is 16 mm.
- Predispense volume is 2 µl.
- Dispensing direction is 'by row'.

Multidrop 384 protocol settings for a 96-well plate:

- Plate is 96 standard (15mm).
- Standard tube (cassette).
- Volume is 20 µl into all columns.
- Speed is 100%.
- X and Y track offsets are 0.
- Dispensing height is 16 mm.
- Predispense volume is 10 µl.
- Dispensing direction is 'by row'.

Multidrop 384 protocol settings for a 384-well plate:

- Plate is 384 standard (15mm).
- Standard tube (cassette).
- Volume is 20 µl into all columns.
- Speed is 100%.
- X and Y track offsets are 0.
- Dispensing height is 16 mm.
- Predispense volume is 10 µl.
- Dispensing direction is 'by row'.

Emulated Command Set

Default protocols

Chapter 5

Using the Commands

When connecting to the instrument, the first command sent must be the VER command. It is recommended that the first command is prefixed with the CR or LF character. These ensure that any garbage characters the instrument may have received before do not interfere with the command. The instrument disconnects from the remote mode when it receives the QIT command. Unlike other commands returning data, the instrument returns the version information even when there is an unacknowledged error. If an error code is returned, the PC should not ignore it.

The following list contains parameters of the default remote protocol, which is loaded when the instrument is started:

Plate is 96 standard (15mm).

Standard tube (cassette).

Volume is 50 µl into all columns.

Speed is 100%.

X and Y dispensing offsets are 0.

Dispensing height is 16 mm.

Predispensing volume is 20 µl.

Dispensing direction is row wise.

The following sequence of commands executes a protocol ("My protocol") from the instrument nonvolatile memory:

BGN 1 My&protocol

Parameter "1" is plate type: 384 wells. Name of the protocol contains a space, which must be replaced using '&' in the command line. The instrument converts the '&' back to space.

It is also possible to 'edit' parameters of the last executed protocol:

Set cassette type with the **SCA** command.

Set plate type with the **SPL** command.

Set column volumes with the **SCV** commands.

Dispense with the **DIS** command.

Appendix A

Plate Definitions

See Figure A–3 and Table A–6 about plate definitions.

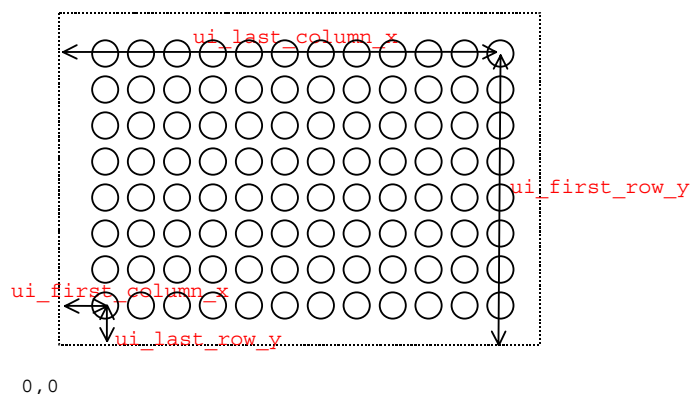


Figure A–3. Plate distance definitions

Table A–6. List of predefined plates

Name	First col. x	Last col. x	First row y	Last row y	Columns	Rows	Plate height	Max. well volume
96 standard (15mm)	14.4 mm	113.4 mm	74.25 mm	11.25 mm	12	8	15.0 mm	500 µl
96 low DW (22mm)	14.4 mm	113.4 mm	74.25 mm	11.25 mm	12	8	22.0 mm	600 µl
96 DW (44mm)	14.4 mm	113.4 mm	74.25 mm	11.25 mm	12	8	44.0 mm	2500 µl
384 low vol. (7.5mm)	12.1 mm	115.6 mm	76.5 mm	9.0 mm	24	16	7.5 mm	25 µl
384 low prof. (10mm)	12.1 mm	115.6 mm	76.5 mm	9.0 mm	24	16	10.0 mm	70 µl
384 standard (15mm)	12.1 mm	115.6 mm	76.5 mm	9.0 mm	24	16	15.0 mm	130 µl
384 DW (22mm)	12.1 mm	115.6 mm	76.5 mm	9.0 mm	24	16	22.0 mm	200 µl
384 DW (44mm)	12.1 mm	115.6 mm	76.5 mm	9.0 mm	24	16	44.0 mm	300 µl
1536 low vol. (5mm)	11.0 mm	116.75 mm	77.6 mm	7.9 mm	48	32	5.0 mm	2.0 µl
1536 std (10.5mm)	11.0 mm	116.75 mm	77.6 mm	7.9 mm	48	32	10.5 mm	13 µl

Notes

[illegible]

[illegible]

Notes

[illegible]

Thermo Fisher Scientific Oy
Microplate Instrumentation
Ratastie 2, P.O. Box 100
FI-01621 Vantaa
Finland

www.thermo.com